Characterization of Message Lengths for Capacity-Achieving Linear PIR

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***Abstract*—We characterize all positive message lengths of capacity-achieving linear private information retrieval (CALPIR) schemes with $M \geq 2$ replicated messages and $N \geq 2$ noncolluding servers. Over every finite field, the achievable lengths are precisely $a(N-1)+bN^{M-1}$, where a, b are nonnegative integers, not both zero. Necessity follows from the pointwise equality conditions of the PIR capacity converse and the integrality of linear conditional entropies. We also revisit T-private PIR for $2 \leq T < N$ and give a pointwise entropy proof that every capacity-achieving linear scheme satisfies $\gcd(N,T)(N/\gcd(N,T))^{M-1} \mid L$. Known constructions attain every positive multiple of this quantity over fields of cardinality at least N. This latter threshold is closely related to the earlier subpacketization results of Zhang and Xu; the pointwise formulation here permits query-dependent answer dimensions.**

I. Problem Setting and Related Work

Let $M, N \geq 2$ and $L \geq 1$ be integers, and let $\mathbb{F}_p$ be a finite field of cardinality p, where p is a prime power. We put $[M] := \{1, \ldots, M\}$ and $[N] := \{1, \ldots, N\}$. All entropies use logarithms to base p. The M messages $W^1, \ldots, W^M \in \mathbb{F}_p^L$ are independent and uniformly distributed, and are replicated at all N servers. The user selects an index θ, uniformly distributed on $[M]$, and draws queries $Q_n \in \mathcal{Q}_n$ for $n \in [N]$, where each query alphabet $\mathcal{Q}_n$ is finite. The answer of server n is A_n. Write $Q_{[N]} = (Q_1, \ldots, Q_N)$, $A_{[N]} = (A_1, \ldots, A_N)$, and $W^{[M]} = ((W^1)^\top, \ldots, (W^M)^\top)^\top$. The requirements are

1) $\mathrm{I}(\theta, Q_{[N]}; W^{[M]}) = 0$ (query independence);
2) $\mathrm{H}(A_n \mid Q_n, W^{[M]}) = 0$ for each n (deterministic answers);
3) $\mathrm{H}(W^m \mid A_{[N]}, Q_{[N]}, \theta = m) = 0$ for every $m \in [M]$ (zero-error correctness);
4) $\Pr(Q_n = q \mid \theta = m) = \Pr(Q_n = q \mid \theta = m')$ for every n, m, m', q (one-server privacy).

A scheme is *linear* if, for every n and $q \in \mathcal{Q}_n$, there is a matrix $\mathbf{G}_n^q$ over $\mathbb{F}_p$ such that $A_n = \mathbf{G}_n^{Q_n} W^{[M]}$. The number of rows of $\mathbf{G}_n^q$ may depend on q; thus the answer is a vector of query-dependent length. We measure the expected *entropy download* by

$$D := \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_n). \tag{1}$$

For a fixed query, deleting linearly dependent answer coordinates preserves the information content and recoverability. Accordingly, in the linear setting D can also be implemented as the expected number of transmitted, linearly independent field symbols, with the query-dependent choice of a row-space basis known to the user. Without this reduction, the actual number of transmitted coordinates may exceed D.

The PIR capacity converse [1] states that

$$\frac{D}{L} \geq \sum_{j=0}^{M-1} N^{-j} = \frac{1 - N^{-M}}{1 - N^{-1}}. \tag{2}$$

We call a linear scheme attaining equality *CALPIR*. Sun and Jafar constructed capacity-achieving schemes with $L = N^{M-1}$ [3]; Tian, Sun, and Chen constructed such schemes with $L = N - 1$ and established the minimum linear message length [2]. Our question is not merely the minimum: we determine *every* positive L for which equality in (2) is possible.

For colluding servers, Zhang and Xu studied the smallest subpacketization of linear capacity-achieving T-private PIR and established a threshold $\gcd(N,T)(N/\gcd(N,T))^{M-1}$ [5]; divisibility arguments already appear in their proof. Xu and Zhang subsequently constructed schemes at the threshold over fields with cardinality at least N [6]. Their fixed-size linear-answer formulation should be distinguished from the query-dependent answer dimensions allowed above. We state the latter model explicitly, without treating the numerical T-private threshold itself as a new discovery.

II. Equality Conditions for Capacity Achievement

The following lemma records the pointwise equality conditions needed below. It also makes explicit how equalities in a standard averaged capacity converse can be passed to each supported query; for $T = 1$ they are the familiar independence and interference properties in [2].

Lemma 1. *Suppose $1 \leq T < N$ and the privacy condition holds for every set of at most T servers. If a PIR scheme attains the T-private capacity, then the following hold for any nonempty $J \subseteq [M]$, any $m \in [M]$, and any $\mathbf{q} = (q_1, \ldots, q_N)$ for which $\Pr(Q_{[N]} = \mathbf{q} \mid \theta = m) > 0$:*

1) *If $m \in J$, the server answers are conditionally independent,*

$$\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m) = \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m). \tag{3}$$

2) *If $m \notin J$, every T-server set spans the full answer information after the indicated messages are revealed: for every $\mathcal{T} \subseteq [N]$ with $|\mathcal{T}| = T$,*

$$\begin{aligned}&\mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m)\\ &\quad = \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^{[M]\setminus J}, \theta = m). \end{aligned} \tag{4}$$

Proof. Set D as in (1). The standard T-private capacity converse [4], [5] may be run for any ordering $m_1, \dots, m_M$ of the messages. It starts with

$$D \geq \mathrm{H}(A_{[N]} \mid Q_{[N]}, \theta = m_1)$$

and, for $S \subsetneq [M]$ nonempty, $m \in S$, and $m' \notin S$, uses

$$\begin{aligned}&\mathrm{H}(A_{[N]} \mid Q_{[N]}, W^S, \theta = m)\\ &\quad \geq \frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}}, W^S, \theta = m)\\ &\quad = \frac{1}{\binom{N}{T}} \sum_{|\mathcal{T}|=T} \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}}, W^S, \theta = m')\\ &\quad \geq \frac{T}{N} \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^S, \theta = m'). \end{aligned} \tag{5}$$

The middle equality follows from T-privacy and the independence of queries and messages. The last step is Han's inequality, applied conditioned on the full query and W^S; each subset entropy is unchanged when the full query is replaced by its local restriction. Correctness inserts exactly L at each successive desired-message step, giving

$$D \geq L \sum_{j=0}^{M-1} (T/N)^j.$$

At capacity all inequalities in this chain are equalities for every permutation: every gap is nonnegative and enters with a positive coefficient.

For $J = [M]$, equality in the initial subadditivity step gives (3). For proper J with $m \in J$, put $S = [M] \setminus J$ and choose a permutation whose first $|S|$ indices are S and whose next index is m. Equality in the Han step on the right of (5), with desired index m, forces equality in conditional Han's inequality. As $T < N$, the equality condition implies mutual independence of the N answers given each supported full query and W^S, proving (3).

For $m \notin J$, put $S = [M] \setminus J$, choose a permutation whose first $|S|$ indices are S, ending in m, and take the next index in J. Equality in the first inequality of (5) yields (4). In both cases, the gaps at a fixed full query are nonnegative: the first is a conditional-entropy monotonicity gap and the second is the conditional Han gap. Since their expectations vanish, they vanish for every supported full query. The same argument applies to every T-subset because the average of its nonnegative gaps vanishes. □

III. Complete Classification for Noncolluding Servers

Define the positive part of the numerical semigroup

$$\mathrm{AM} := \{a(N-1) + bN^{M-1} : a, b \in \mathbb{Z}_{\geq 0}\} \setminus \{0\}. \tag{6}$$

Theorem 1. *For any finite field $\mathbb{F}_p$ and any integers $M, N \geq 2$, a CALPIR scheme with positive message length L exists if and only if $L \in \mathrm{AM}$.*

Proof. The constructions with message lengths $N-1$ and N^{M-1} exist over every finite field [2], [3]. Direct sums preserve linearity, correctness, privacy, and equality in the normalized download bound, proving sufficiency.

For necessity fix $m \in [M]$ and a supported query vector $\mathbf{q}$. Write

$$h(\mathbf{q}) := \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^m, \theta = m).$$

Take $T = 1$. By Lemma 1 with $J = [M]$ and $J = [M] \setminus \{m\}$, respectively, and by correctness,

$$\begin{aligned} N h(\mathbf{q}) &= \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_{[N]} = \mathbf{q}, W^m, \theta = m)\\ &\leq \sum_{n \in [N]} \mathrm{H}(A_n \mid Q_{[N]} = \mathbf{q}, \theta = m)\\ &= \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, \theta = m)\\ &= L + h(\mathbf{q}). \end{aligned} \tag{7}$$

The last identity is the chain rule using the zero-error recovery of W^m and query independence. Since the scheme is linear, $h(\mathbf{q})$ is the rank of the answer map restricted to the $M-1$ undesired message blocks and hence is an integer. Thus

$$h(\mathbf{q}) \leq \left\lfloor \frac{L}{N-1} \right\rfloor. \tag{8}$$

Averaging over supported queries and using the equality $D = \mathrm{H}(A_{[N]} \mid Q_{[N]}, \theta = m)$ from the converse,

$$\begin{aligned} D - L &= \mathrm{H}(A_{[N]} \mid Q_{[N]}, W^m, \theta = m)\\ &\leq \left\lfloor \frac{L}{N-1} \right\rfloor. \end{aligned} \tag{9}$$

Capacity attainment and (2) give

$$L - \frac{L}{N^{M-1}} \leq (N-1) \left\lfloor \frac{L}{N-1} \right\rfloor. \tag{10}$$

Let b be the least nonnegative residue of L modulo $N-1$. Then (10) is equivalent to $bN^{M-1} \leq L$. Since $N^{M-1} \equiv 1 \pmod{N-1}$,

$$a := \frac{L - bN^{M-1}}{N-1} \in \mathbb{Z}_{\geq 0}.$$

Consequently $L = a(N-1) + bN^{M-1} \in \mathrm{AM}$. □

A. Consequences of the Classification

For $N = 2$, $\mathrm{AM} = \mathbb{Z}_{\geq 1}$. If $N \geq 3$, the two semigroup generators are coprime. The Frobenius coin theorem then shows that the largest positive length outside AM is

$$(N-2)(N^{M-1} - 1) - 1, \tag{11}$$

except when $(M, N) = (2, 3)$, for which (11) still equals 1 but the interval considered below is empty. There are exactly

$$\frac{(N-2)(N^{M-1}-1)}{2} \tag{12}$$

nonachievable positive lengths, and every $L \geq (N-2)(N^{M-1}-1)$ is achievable. For $N \geq 3$ and $(M,N) \neq (2,3)$, the number of achievable lengths in $[N-1,(N-2)(N^{M-1}-1)-1]$ is

$$\frac{(N-2)(N^{M-1}-1)}{2}-1. \tag{13}$$

For $(M,N)=(2,3)$, $\mathrm{AM}=\mathbb{Z}_{\geq 2}$.

IV. Extension to T-Private PIR

Suppose $2 \leq T < N$. Privacy is now required for every $\mathcal{T} \subseteq [N]$ with $|\mathcal{T}| \leq T$:

$$\begin{aligned}&\Pr(Q_{\mathcal{T}} = q_{\mathcal{T}} \mid \theta = m)\\ &\quad = \Pr(Q_{\mathcal{T}} = q_{\mathcal{T}} \mid \theta = m')\end{aligned} \tag{14}$$

for all $m, m' \in [M]$ and all $q_{\mathcal{T}}$. The capacity bound is [4]

$$\frac{D}{L} \geq \sum_{j=0}^{M-1} (T/N)^j. \tag{15}$$

Theorem 2. *Every capacity-achieving linear T-private PIR scheme over any finite field $\mathbb{F}_p$ has a message length satisfying*

$$\gcd(N,T)\left(\frac{N}{\gcd(N,T)}\right)^{M-1} \mid L. \tag{16}$$

Conversely, for every $p \geq N$ and every positive L satisfying (16)*, such a scheme exists over $\mathbb{F}_p$.*

The numerical threshold in (16) is known from [5], [6]. The argument below emphasizes that the same divisibility condition is necessary in our query-dependent answer-dimension model, using pointwise conditional entropy equalities rather than an assumption of a fixed answer length at each server.

Lemma 2. *Fix a nonempty proper $J \subsetneq [M]$. For every $m \in [M]$ and every $\mathbf{q}$ supported given $\theta = m$, the values*

$$\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m), \quad n \in [N], \tag{17}$$

are equal.

Proof. Put $r_n(q_n) := \mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J})$. Query independence implies that this quantity is independent of the requested index when q_n is held fixed. Choose $m' \in J$ and $m'' \notin J$. For any supported $\mathbf{q}'$ given $\theta = m''$ and any T-subset $\mathcal{T}$, privacy supplies a query vector supported for $\theta = m'$ with the same restriction $q'_{\mathcal{T}}$. By the two parts of Lemma 1,

$$\begin{aligned}\sum_{n\in\mathcal{T}} r_n(q'_n) = \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}',\\ W^{[M]\setminus J}, \theta = m'').\end{aligned} \tag{18}$$

The right-hand side is independent of $\mathcal{T}$. Comparing two T-subsets differing in one server index shows that all $r_n(q'_n)$ are equal because $T < N$. Finally, given an arbitrary supported $\mathbf{q}$ and distinct n, n', a T-subset containing n, n' exists since $T \geq 2$. Privacy supplies $\mathbf{q}'$ supported for $\theta = m''$ with the same local query values at those two servers. Hence $r_n(q_n) = r_{n'}(q_{n'})$. □

Proof of Theorem 2. Let $1 \leq s \leq M-1$, choose $J \subseteq [M]$ with $|J| = s$, and $m \in J$. For any supported full query $\mathbf{q}$ we prove, by induction on s, that each server has conditional entropy

$$\begin{aligned}&\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m)\\ &\quad = \frac{L}{N^s}\sum_{j=0}^{s-1} N^{s-1-j}T^j.\end{aligned} \tag{19}$$

For $s = 1$, conditioning on every message except the desired one and using correctness yields total conditional entropy L at each supported full query. Equation (3) and Lemma 2 then give L/N per server.

For $s \geq 2$, select $m' \in J \setminus \{m\}$, a T-subset $\mathcal{T}$, and a supported query $\mathbf{q}'$ given $\theta = m'$ with $q'_{\mathcal{T}} = q_{\mathcal{T}}$, as guaranteed by privacy. Using correctness, the two parts of Lemma 1, and the induction hypothesis, we obtain

$$\begin{aligned}&N\mathrm{H}(A_n \mid Q_n = q_n, W^{[M]\setminus J}, \theta = m)\\ &= \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus J}, \theta = m)\\ &= L + \mathrm{H}(A_{[N]} \mid Q_{[N]} = \mathbf{q}, W^{[M]\setminus(J\setminus\{m\})}, \theta = m)\\ &= L + \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q_{\mathcal{T}}, W^{[M]\setminus(J\setminus\{m\})}, \theta = m)\\ &= L + \mathrm{H}(A_{\mathcal{T}} \mid Q_{\mathcal{T}} = q'_{\mathcal{T}}, W^{[M]\setminus(J\setminus\{m\})}, \theta = m')\\ &= L + \frac{TL}{N^{s-1}}\sum_{j=0}^{s-2} N^{s-2-j}T^j.\end{aligned} \tag{20}$$

This proves (19). The left-hand side of (19) is the rank of a fixed linear map over $\mathbb{F}_p$ and therefore an integer. In particular, $s = 1$ gives $N \mid L$. Taking $s = M-1$ gives

$$\frac{L}{N^{M-1}}\sum_{j=0}^{M-2} N^{M-2-j}T^j \in \mathbb{Z}. \tag{21}$$

Since $N \mid L$ and

$$\begin{aligned}\gcd\Bigg(&\left(\frac{N}{\gcd(N,T)}\right)^{M-2},\\ &\sum_{j=0}^{M-2}\left(\frac{N}{\gcd(N,T)}\right)^{M-2-j}\left(\frac{T}{\gcd(N,T)}\right)^{j}\Bigg) = 1,\end{aligned} \tag{22}$$

(21) implies $N(N/\gcd(N,T))^{M-2} \mid L$, which is (16).

For sufficiency, Xu and Zhang constructed a linear capacity-achieving T-private scheme with length equal to the left-hand divisor of (16) over every finite field with $p \geq N$ [6]. Direct sums yield every positive multiple, without changing the field. □

For $p < N$, scalar extension yields a more limited sufficient condition. If $p^s \geq N$, a scheme over $\mathbb{F}_{p^s}$ can be viewed as a scheme over $\mathbb{F}_p$ after expressing each field symbol in an $\mathbb{F}_p$-basis. This proves achievability for positive lengths divisible by

$$s\gcd(N,T)\left(\frac{N}{\gcd(N,T)}\right)^{M-1}.$$

The sufficiency of (16) itself over all smaller fields is not asserted here.

V. Conclusion

For noncolluding servers, the set of all positive message lengths of capacity-achieving linear PIR is the numerical semigroup generated by $N-1$ and N^{M-1}, with no restriction on the finite-field size. For $2 \leq T < N$, the familiar subpacketization divisibility condition continues to hold when answer dimensions depend on the query, under the entropy-download convention; matching constructions are available when the field cardinality is at least N. The two cases underscore how noncolluding PIR can admit asymmetric short-message constructions that do not persist under stronger collusion privacy.

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